



An integrated approach of vortex-induced vibration for long-span bridge with inhomogeneous cross-sections

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ABSTRACT

Conventional models resolve the vortex-induced vibration (VIV) problem for long-span bridge with an essential assumption that the cross-section is uniform. When the cross-sections are altered in span-wise direction, i.e., inhomogeneous cross-section design, conventional approaches are inapplicable, and the global VIV becomes ambiguous and difficult to evaluate directly. This paper presents an integrated approach which combines the Scanlan's nonlinear VIV model and an aerodynamic damping model to characterize the global VIV for long-span bridge with inhomogeneous cross-section design. In the proposed approach, the Scanlan's nonlinear model is utilized to characterize the vortex-induced motivating force on cross-sections at lock-in whereas the aerodynamic damping model is employed to describe the air-induced hindering effect on cross-sections away from lock-in. The parameters in the integrated model are identified using either a 'decay-to-resonance' method or free decay vibration test via sectional model tests. The proposed approach is validated through a proof-of-concept wind tunnel experiment. An engineering practice of an urban rail-road long-span bridge with span-wise inhomogeneous cross-section design is demonstrated and discussed as an example. The VIV performance with and without considering inhomogeneous effects are evaluated and compared. The effects of various section deployment plans as well as their global performance are investigated and discussed.

1. Introduction

Vortex-induced vibration (VIV) for long-span bridge has drawn extensive attention in wind engineering community in recent decades due to its wide impacts on driving safety and structural durability. Though VIV has an inherent nature of self-limited oscillations in amplitude, it might cause unexpected interruptions in traffic operations and threatens the driving safety, as reported in some operational bridges such as Trans-Tokyo Bay Bridge (2002), Volgograd Bridge (2012), Jindo Bridge (2013), Xihoumen Bridge (2014b). Evaluation of VIV performance becomes one of the most foundational tasks in the preliminary design and maintenance for long-span bridge.

Besides the basic purpose to provide a passage over physical obstacles, modern bridge tends to design utilizing comprehensive concepts to enable the structure with more functions such as environmental-friendly and harmonious coexistence of humans and infrastructures, which may increase the complexity in structures, e.g., span-wise inhomogeneous sections, and leads to new challenges in wind-resistance design (Honda et al., 1992; Li et al., 2013; Lü et al., 2018; Xiong et al., 2020). A typical

case demonstrated and discussed in this study is a long-span rail-road urban bridge designed with two different cross-sections in span-wise direction, i.e., a fully enclosed section enhanced with noise barrier stretching from bridge ends to 1/3 of the span and a regular truss section in the middle, as shown in Fig. 1. Such an inhomogeneous design aims to take advantage of the superiority of enclosed noise barrier in noise cancelling to protect urban inhabitants from noise pollution near two bridge ends while a regular truss section is designed in the middle to reduce financial expenses since it is unnecessary to consider noise pollution from a far distance. Such a span-wise inhomogeneous design could provide the bridge with more functions such as noise-friendly to urban inhabitants whereas it raises other issues from the perspective of wind-resistance design, e.g., VIV evaluation and mitigation. In the inhomogeneous cross-section design, cross-sections may have significantly different lock-in wind speeds and aerodynamic behaviors (Bahramiasl and de Langre, 2021), e.g., one section may meet the necessary flow condition to trigger lock-in while others are away from lock-in wind speeds or have different VIV force, and thus the global VIV behavior of the entire bridge become ambiguous and difficult to evaluate directly.

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Conventional VIV approaches aim to model the vortex-induced behaviors of the bridge with an essential assumption that the cross-sections and their aerodynamic characteristics are span-wise uniform (Wardlaw, 1980) and have the identical lock-in wind speeds, i.e., homogeneous design. The span-wise inhomogeneity in aerodynamics makes conventional approaches inapplicable. To sum up, VIV evaluation for long-span bridge with inhomogeneous sections remains an open issue and requires specific investigations.

Extensive efforts have been devoted to model vortex-induced force and motions, either analytically or numerically. Van der Pol's oscillator model (Kanamaru, 2007; Van der Pol, 1920) is a foundational approach to establish the vortex-induced behaviors of structures and several semi-empirical mathematical models were proposed at the basis of it (Balasubramanian and Skop, 1996; Facchinetti et al., 2002, 2004; Marra et al., 2011; Scanlan, 1981; Srinil and Zanganeh, 2012). One pioneering work by Ehsan and Scanlan (1990) established a nonlinear force model to predict the amplitude-dependent aerodynamic force, in which the parameters were identified by 'decay-to-resonance' or 'growth-to-resonance' vibration tests in wind tunnel, and the model is widely employed by wind engineering community. Chen (2014) presented a nonlinear model in the form of polynomial function of time-varying velocity and displacement of vibration. Most of the vortex-induced approaches aim to model the vortex-induced behaviors of the bridge with an essential assumption that the cross-sections are span-wise uniform and becomes inapplicable for the span-wise inhomogeneity problem. More studies are needed to investigate the vortex-induced vibration for long-span bridges with inhomogeneous cross-section design.

Wind tunnel test remains one of the most feasible ways to investigate the vortex-induced behaviors for long-span bridges, either a 2-dimensional sectional model test or a 3-dimensional full bridge test (Cermak, 2003; Irwin, 1992; Nieto et al., 2009). With respect to inhomogeneous design, since the effect of span-wise inhomogeneous aerodynamics plays a prominent role, a full bridge test is by instinct the most straightforward and sensible to take the variation of cross-sections into account, but the drawbacks are also obvious that it is expensive in terms of both finance and labor and more importantly, its accuracy may inevitably reduce due to the over-scaled testing model (Frandsen, 2001; Hjorth-Hansen, 2017; Larsen et al., 2000; Li et al., 2014a; Nieto et al., 2009; Wardlaw, 1980). A sectional wind tunnel model test is the most practical way to evaluate the VIV performance and aerodynamic parameters, of which results can also be extended to a full bridge scenario associated with the VIV model considering structural mode shape (Zhang et al., 2014). By performing sectional model test, Zhu et al. (2017) presented a simplified nonlinear model to characterize the vertical vortex-induced force on box girder to predict the stable amplitude

of VIV. Xu et al. (2020) compared the feasibility of some commonly used VIV models through sectional model test and emphasized the prominent role of damping in VIV investigation. Compared with a full-bridge test, a sectional wind tunnel test shows significant benefits in expense, flexibility, and accuracy. The presented study follows such a fashion with aims to establish an approach only using 2-dimensional section testing instead of a high-cost full bridge test to approximate the global VIV performance considering inhomogeneous sections.

This paper presents an integrated approach which combines the Scanlan semi-empirical nonlinear VIV model and an aerodynamic damping model to characterize the global VIV behaviors for the entire bridge with inhomogeneous cross-section design. The parameters in the integrated model are identified using either a 'decay-to-resonance' method or free vibration method via 2-dimensional wind tunnel tests. The proposed approach is validated through a proof-of-concept experiment in which the VIV performance of a rigid bridge segment with two different cross-sections design is investigated and compared with experimental data. An engineering practice of an urban rail-road long-span bridge with span-wise inhomogeneous cross-sections design is demonstrated and discussed as an example. The inhomogeneous cross-sections are composed of a noise-friendly fully enclosed section which is deployed at bridge ends to eliminate noise pollution to urban habitants while a regular truss section is applied in the central area of the span to reduce unnecessary costs. A series of wind tunnel tests are performed to identify the aerodynamic parameters as well as vortex-induced characteristics of the sections. The VIV performance with and without considering inhomogeneous effects are evaluated and compared. The effects of various section deployment plans as well as their global performance are investigated and discussed. Some conclusions are summarized at the end.

2. Integrated vortex-induced force model

2.1. Fundamental formulation

Regarding to the foundational problem for long-span bridge with inhomogeneous sections design, without loss of generality, a typical scenario with two different cross-sections, i.e., cross-section 1 and cross-section 2 (hereinafter referred as CS1 and CS2) is demonstrated in Fig. 2.

Assume that at incoming flow speed U , steady-state VIV is undergoing in the entire bridge at the foundational vertical structural frequency f . Due to the inhomogeneous cross-section design, girder segment with CS1 meets the sufficient flow condition to induce VIV, i.e., flow speed U is within the lock-in and drives the oscillation of the entire bridge whereas flow speed U is not within the lock-in for girder segment

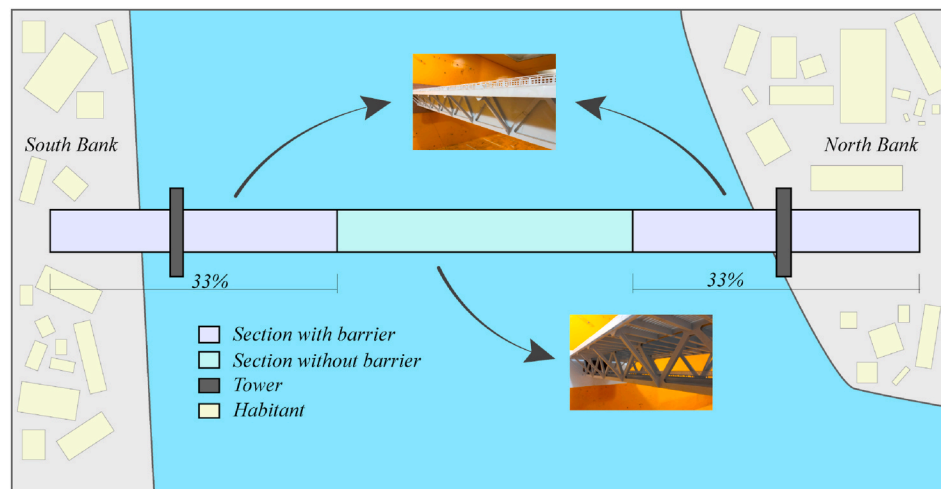


Fig. 1. A typical bridge with inhomogeneous cross-section enhanced with enclosed noise barrier.

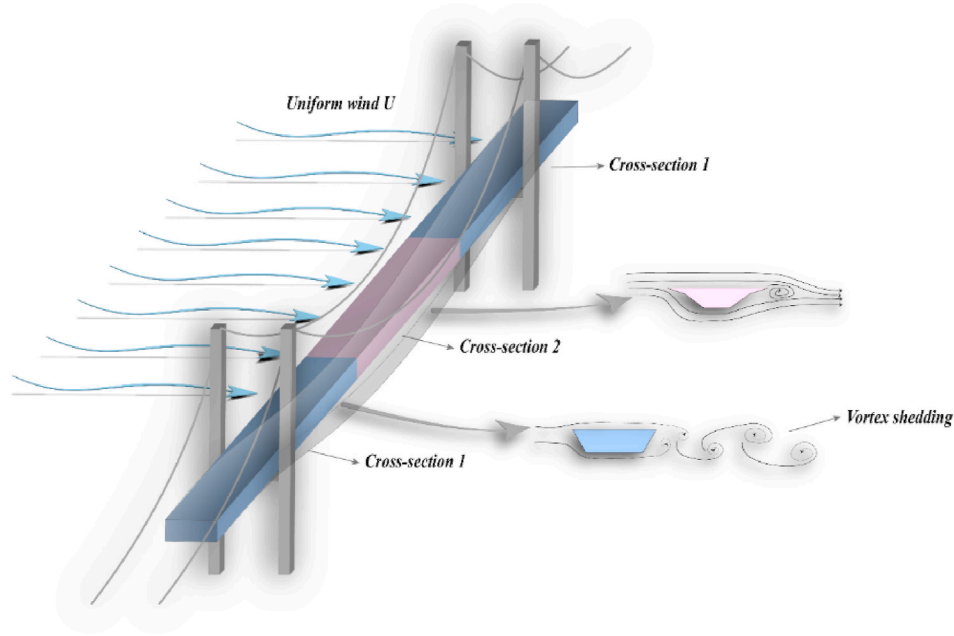


Fig. 2. A typical scenario with two different cross-sections showing the fundamental problem of bridge VIV with inhomogeneous cross-sections.

with CS2, and thus CS2 is vibrating passively due to the motivation of CS1. Note that CS2 may also be within the lock-in according to its Strouhal number, the assumption that CS2 is vibrating passively is a typical case to clarify the fundamental problem and illustrate the integrated model. Under such a circumstance, the bridge's motion could be expressed as:

$$[\mathbf{m}][\ddot{\mathbf{y}}] + [\mathbf{c}][\dot{\mathbf{y}}] + [\mathbf{k}][\mathbf{y}] = [\mathbf{F}] \quad (1)$$

where $[\mathbf{m}]$ is the mass matrix, $[\mathbf{c}]$ is the damping matrix, $[\mathbf{k}]$ is the stiffness matrix, $\mathbf{y}$ is the vertical displacement vector of the structure, and $[\mathbf{F}]$ is the aerodynamic force vector. The force vector $[\mathbf{F}]$ on the right-side of the equation can be further written as:

$$[F_v \quad F_m] \quad (2)$$

in which F_v is the positive vortex-induced force vector on segments with CS1 and F_m is the negative motion-induced aerodynamic force vector, i.e., aerodynamic damping acting on the segments with CS2. Note that the spatial effect on the variation of aerodynamic forces at the transition point between CS1 and CS2 are reasonably ignored to reduce the complexity of the problem. The vortex-induced force F_v can be described using Scanlan's semi-empirical non-linear model (Ehsan and Scanlan, 1990), which reads:

$$F_v = \frac{1}{2} \rho U^2 (2D) \left[Y_1(K) \left(1 - \varepsilon(K) \frac{y^2}{D^2} \right) \frac{\dot{y}}{U} + Y_2(K) \frac{y}{D} + \frac{1}{2} C_L \sin(\omega_s t + \theta) \right] \quad (3)$$

in which ρ is air density; U is the wind velocity incoming; D is the across-flow dimension of the section; ω_s is vortex shedding frequency; ω is oscillation frequency and $\omega = \omega_s$ when VIV occurs; K is the non-dimensional reduced frequency which reads $K = \omega D/U$; C_L is transient lift coefficient. $Y_1(K)$, $\varepsilon(K)$, $Y_2(K)$, C_L are aerodynamic parameters needed to be determined by wind tunnel test. In Eq. (3), $Y_1(K)$ and $Y_2(K)$ are the aerodynamic damping and stiffness term, respectively, associated with the structure's acceleration and motion and they compose the self-excitation force. $\varepsilon(K)$ represents the self-limiting characteristics of vortex-induced vibration. At lock-in, Y_2 and C_L are much smaller compared to the aerodynamic damping term Y_1 (Sun et al., 2014) and they can be reasonably ignored. Thus Eq. (3) yields:

$$F_v = \rho U^2 D \left[Y_1(K) \left(1 - \varepsilon(K) \frac{y^2}{D^2} \right) \frac{\dot{y}}{U} \right] \quad (4)$$

Since CS2 is vibrating passively due to the motivation of CS1, the motion-induced aerodynamic force, i.e., aerodynamic damping acting on CS2 must be properly considered since VIV is sensitive to damping. By imposing a linear aerodynamic damping model, the motion-induced force on CS2 can be expressed as:

$$F_m = \rho U D H_1(K) \dot{y} \quad (5)$$

in which K is reduced frequency which reads $K = \omega D/U$ and ω can be considered as natural frequency with the ignoring of aerodynamic stiffness; H_1 is the aerodynamic damping parameter of CS2 and could be identified via wind tunnel test. Among the to-be-identified aerodynamic parameters, Y_1 is positive, which drives the structure's motion whereas H_1 is negative, increasing the overall damping and hindering the structure's motion.

When the vortex-shedding frequency approaches the natural frequency, the bridge's vertical motion is expected to occur at or near the natural frequency if the structural eigenfrequencies are well separated (Ehsan and Scanlan, 1990). By introducing the generalized coordinates, the vertical motion $\mathbf{y}$ of the bridge at the span-wise location $\mathbf{x}$ at any time t can be non-dimensional and multiplied by the mode shape function to obtain the following form:

$$\mathbf{y}(\mathbf{x}, t) = \varphi(\mathbf{x}) \xi(t) D \quad (6)$$

or in matrix form:

$$[\mathbf{y}] = D[\varphi] \xi(t) \quad (7)$$

where $\varphi(\mathbf{x})$ is the mode shape function, $[\varphi]$ is the mode shape vector, $\xi(t)$ is the generalized coordinate. The following equation can be obtained by inserting Eq. (7) into Eq. (1) and multiplying both sides by $[\varphi]$ as:

$$M \left[\ddot{\xi}(t) + 2\zeta \omega \dot{\xi}(t) + \omega^2 \xi(t) \right] = \frac{1}{D} [\varphi]^T [\mathbf{F}] \quad (8)$$

$[\mathbf{F}]$ can be expanded as:

$$F_v = \rho U D^2 Y_1(K) \varphi \left[1 - \varepsilon(K) \varphi^2 \xi^2(t) \right] \dot{\xi}(t) \quad (9)$$

$$F_m = \rho U D^2 H_1(K) \varphi \dot{\xi}(t) \quad (10)$$

where M is the generalized mass $M = [\varphi]^T [m] [\varphi]$, ζ is the damping ratio. According to the energy conservation, when the bridge oscillates at steady lock-in stage, the energy dissipated by the mechanical damping is equal to the input energy from the flow. In addition, it is reasonable to assume that the generalized coordinates are in harmonic form, i.e., $\xi(t) = \xi_0 \cos(\omega t)$. This assumption associated with energy conservation condition leads to the following solution:

$$\xi_0 = 2 \left[\frac{[\varphi]^T [\mathbf{C}_1 \varphi] - \frac{2nM\zeta K}{\rho D^2 L}}{[\varphi]^T [\mathbf{C}_1 \varphi^3 \mathbf{C}_2]} \right]^{1/2} \quad (11)$$

in which, n is the number of elements divided along the span; L is the length of girder. $[\mathbf{C}_1]$ and $[\mathbf{C}_2]$ are aerodynamic parameters vectors. Since the bridge has two different cross-sections, $[\mathbf{C}_1]$ is composed of two constants Y_1 and H_1 , and $[\mathbf{C}_2]$ is composed of ε and zeros. $[\mathbf{C}_1]$ and $[\mathbf{C}_2]$ can be written as:

$$\begin{aligned} [\mathbf{C}_1] &= [A_{1,1} \ A_{1,2} \ \dots \ A_{1,i} \ \dots \ A_{1,n}] \\ [\mathbf{C}_2] &= [A_{2,1} \ A_{2,2} \ \dots \ A_{2,i} \ \dots \ A_{2,n}] \end{aligned} \quad (12)$$

$A_{1,i}, A_{2,i}$ are aerodynamic parameters of the i_{th} element under incident flow speed U and have the following relationship depending on whether the corresponding cross-section is at lock-in or not:

$$\begin{aligned} \text{if the cross-section of } i_{th} \text{ element is at lock-in:} \\ A_{1,i} &= Y_{1,i} \\ A_{2,i} &= \varepsilon_i \\ \text{if not:} \\ A_{1,i} &= H_{1,i} \\ A_{2,i} &= 0 \end{aligned} \quad (13)$$

Note that if there are more than two cross-sections along the span, $[\mathbf{C}_1]$ and $[\mathbf{C}_2]$ can be obtained by combining the aerodynamic parameters of each cross-section according to the actual span-wise cross-section design. As a typical case, for a sectional wind tunnel test of rigid bridge model, $[\varphi]$ is a unit vector and $[\varphi] = I$. Thus Eq. (11) becomes:

$$\xi_0 = 2 \left[\frac{[\mathbf{C}_1]^T I - \frac{2nM\zeta K}{\rho D^2 L}}{[\mathbf{C}_1]^T [\mathbf{C}_2]} \right]^{1/2} \quad (14)$$

If the aerodynamic parameters are constant along the span, i.e., the bridge has only a single cross-section, Eq. (14) can be further reduced to:

$$\xi_0 = 2 \left[\frac{Y_1 I - \frac{2nM\zeta K}{\rho D^2 L}}{Y_1 \varepsilon} \right]^{1/2} \quad (15)$$

Eqs. (14) and (15) can be applied to the validation test.

2.2. Parameters identification

The parameters Y_1 , ε and H_1 in the proposed approach which characterizes the aerodynamic behaviors of bridge cross-sections need to be identified via wind tunnel tests. The aerodynamic parameters Y_1 , ε which represents the lock-in characteristics are identified through a so-called decay-to-resonance test whereas the aerodynamic damping parameter H_1 out of lock-in is carried out via a free vibration test.

2.2.1. Decay-to-resonance method

A pioneering work by Ehsan and Scanlan (1990) used the decay-to-resonance test to identify the lock-in aerodynamic parameters Y_1 and ε . In such a test, a displacement larger than the steady-state amplitude is set to the model and then released. The amplitude of the model will decay over time and eventually converge to the steady-state

amplitude. The time series of displacements, as shown in Fig. 3, will be recorded, and used for the identification.

In Fig. 3, η is the non-dimensional cross-flow displacement, $\eta = y/D$, and s is the non-dimensional time, $s = Ut/D$. A_0 is the non-dimensional initial displacement deployed to the sectional model; β is the non-dimensional steady-state amplitude. For parameters Y_1 and ε , the identification is based on the decimal transformation and linear fitting. Y_1 and ε can be expressed as:

$$Y_1 = \frac{1}{4m_r} [\alpha \beta^2 + 8\zeta K] \quad (16)$$

$$\varepsilon = \frac{\alpha}{m_r Y_1} \quad (17)$$

$$m_r = \frac{\rho D^2}{m} \quad (18)$$

where α is the parameter which needs to be determined, m_r is the non-dimensional mass ratio, the decimal transformation and linear fitting are written as follows:

$$F = \ln \left\{ \frac{A_0^2 [A^2(s) - \beta^2]}{A^2(s) (A_0^2 - \beta^2)} \right\} \quad (19)$$

$$X = s \quad (20)$$

$$F = aX \quad (21)$$

where $A(s)$ is a set of peak amplitudes in the decay data. F shall be determined associated with A_0 , $A(s)$, and β from the decay curve. X equals the non-dimensional time s , and a linear fitting can be conducted with Eq. (22). The slope of the line a can be written as:

$$a = -\frac{\alpha \beta^2}{4} \quad (22)$$

With Eq. (22), α can be obtained, and Eqs. (16) and (17) can be solved.

2.2.2. Free decay vibration test

This method is applied to identify aerodynamic damping parameter H_1 when VIV does not occur on the testing model (Cao and Ge, 2017). In a similar fashion, an initial displacement is deployed and released on the testing model at a certain wind speed to proceed a free vibration test. Since no VIV occurs, the displacement of the model will decay to zero over time. A typical decay process is shown in Fig. 4.

The displacement of the model is essentially the free vibration of an oscillator with damping, which can be expressed as:

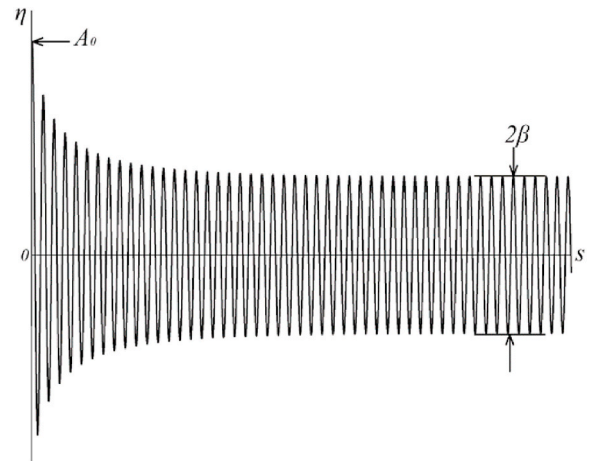


Fig. 3. Decay-to-Resonance curve.

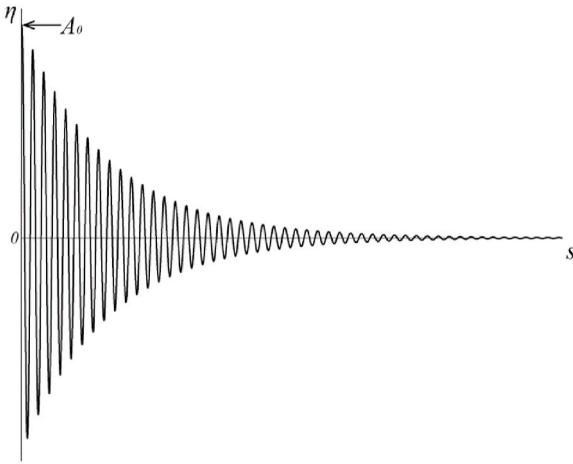


Fig. 4. Process of free decay.

$$\eta = A_0 e^{-\gamma K s} \sin(Ks + \varphi) \quad (23)$$

$$A(s) = A_0 e^{-\gamma K s} \quad (24)$$

In a similar fashion as the Decay-to-Resonance Method, Eq. (23) can be rewritten as follows by the decimal transformation:

$$F = aX + b \quad (25)$$

$$F = \ln[A(s)] \quad (26)$$

$$\begin{aligned} a &= -\gamma K \\ b &= \ln(A_0) \end{aligned} \quad (27)$$

in which, γ is the damping ratio of the oscillator, a is the slope of the line used to be fitting, and b is the intercept.

Based on the decay curve, a set of peak amplitude $A(s)$ can be obtained, and the initial displacement A_0 is known. Therefore, the linear fitting can be performed employing Eq. (25)–(27), and the slope a can be obtained. Parameter H_1 can be determined as:

$$H_1 = -2K(\gamma - \zeta) / m_r \quad (28)$$

Once the aerodynamic parameters are determined, the proposed model is readily to deploy on prediction of VIV. To sum up, the proposed model can be applied to investigate the VIV of long-span bridge with inhomogeneous cross-sections following the procedure:

- (1) Investigate the interested eigenfrequencies and corresponding mode shape of the bridge.
- (2) Conduct wind tunnel tests for each cross-sections using spring-support sectional models. For each cross-section, using decay-to-resonance test to identify aerodynamic parameters Y_1 and ε within vortex lock-in wind speeds when VIV occurs whereas using decay data from free vibration test to identify aerodynamic parameter H_1 at wind speeds away from lock-in range;
- (3) Combine the identified parameters to construct matrix $[C_1]$ and $[C_2]$ and the global VIV can be calculated using Eq. (11) and Eq. (7).

A flow chart of the procedure is illustrated in Fig. 5:

3. Proof-of-concept case study

3.1. Description of example case

An engineering practice of a long-span suspension bridge is discussed in this study as an example. As shown in Fig. 6(a), the example bridge

has a main span of 720 m and two side spans of 75m. The steel truss girder is composed of double decks which the upper deck serves the road traffic while the lower one serves the urban rail lines. To protect the urban habitants near bridge ends from operational noise by railways, a fully enclosed noise barrier stretched from both bridge ends to 1/3 length of the bridge are installed to create a segregated space for trains and thus could reduce the noise to a considerable degree. Near the middle of the span, a regular truss girder without barrier is employed. The deployment of bridge sections is shown in Fig. 1. Since VIV is by theory more easily to occur at the fundamental frequency (Fujino and Yoshida, 2002; Owen et al., 1996; Smith, 1980), only the first vertical mode shape is considered in this study, as shown in Fig. 6(b) and (c).

The inhomogeneous cross-section design could improve the harmony between infrastructure and inhabitants whereas it raises challenges on global VIV evaluation. Fig. 7 demonstrates the layout of cross-sections as well as the sketch of testing models with and without noise barrier. For sake of conciseness, the cross-section with and without barrier is denoted as CS1 and CS2 hereinafter, respectively. It can be observed that compared with the original truss section without barrier (CS2) shown in Fig. 7(c) and (d), the installation of the fully enclosed noise barrier (CS1) in Fig. 7(a) and (b) significantly lowers the ventilation of the girder and turns the original truss girder into a box-like one. Such a modification on cross-section may apparently degrade the wind-resistance performance of the girder and needs to investigate carefully.

3.2. Experimental configuration

The rigid sectional model wind tunnel tests with elastic suspension are performed to evaluate the VIV performance and identify the aerodynamic parameters of each cross-section. The wind tunnel test was conducted at the XNJD-1 wind tunnel of Southwest Jiaotong University in China. The wind tunnel is 2 m in height and 2.4 m in width. The testing model of bridge girder with a scale ratio of 1/64.4 and 9 truss panels is designed. Fig. 8(a) and (b) portray the spring-supported section models with and without barrier. The tests are performed under uniform smooth flows at angle of attack of -3° , 0° and $+3^\circ$ (nose-up positive) and for sake of conciseness, only the results of $+3^\circ$ will be discussed as a representative case in the following contents because the VIV under $+3^\circ$ is observed to be the most remarkable. The mass per unit length of the testing model are 14.060 kg/m. The vertical frequency of the model is 2.888 Hz. The vertical damping ratio is set at 0.48%. The testing wind speed ranges from 0 to 7 m/s with a flexible increment step of 0.05–0.13 m/s to record vibration data with a refined wind speed resolution, especially near lock-in speeds. When VIV is undergoing on the testing model, a decay-to-resonance test is carried out to obtain aerodynamic parameters Y_1 and ε for the cross-section whereas a free vibration test is carried out to obtain aerodynamic parameter H_1 when VIV is absent on the model.

3.3. Validation test

A validation test is performed to verify the feasibility of the proposed model on prediction of VIV for bridge with inhomogeneous cross-sections. A full bridge wind tunnel test is by instinct the most straightforward way to validate the inhomogeneous problem. However, considering the inevitable drawbacks and high costs of a full bridge test, a rigid spring-supported sectional model is designed as an alternative to validate the proposed approach. The validation model is composed of a regular truss segment in the middle and two fully enclosed segments with noise barrier on both sides which stretches from both ends to 1/3 length of the model. The validation model is illustrated in Fig. 9. Compared with a full bridge test, such a rigid model design can also reflect the essential mechanics of VIV with inhomogeneous cross-sections only with constant mode shape $\varphi(x) = 1$ instead of a variable mode shape of a full bridge. All the experimental configurations remain the same as introduced in Section 3.2. The variation of VIV amplitude

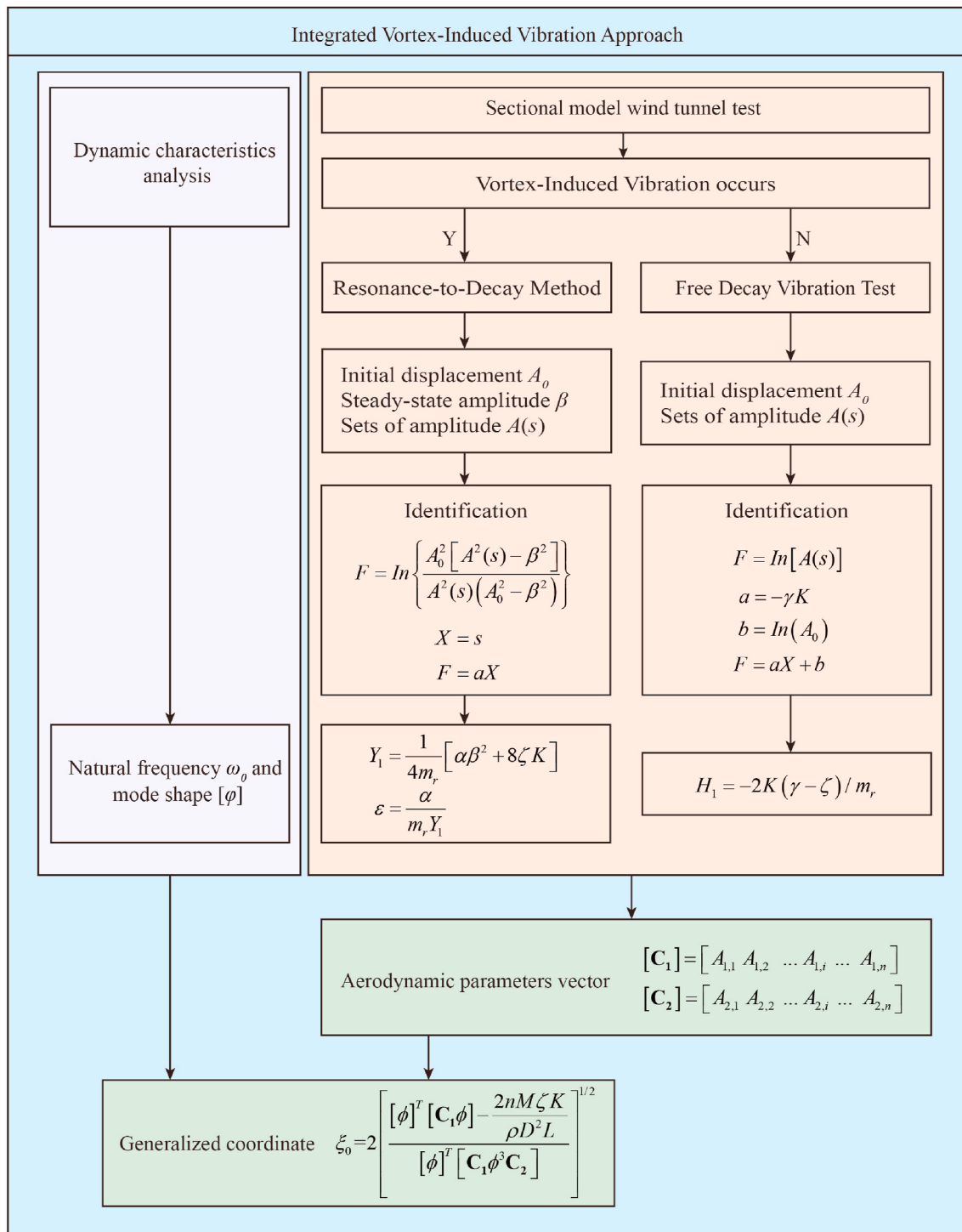


Fig. 5. Flow chart using the proposed model for VIV prediction.

against wind speed is tested and recorded. As the aerodynamic parameters for both cross-sections, i.e., the truss section and the fully enclosed section with barrier are identified in Section 3.2 and are readily to applied in the proposed model, the VIV amplitude under different wind speeds can be predicted and compared against experimental data.

4. Results and discussion

4.1. Experimental results

The VIV amplitude against wind speeds of CS1 and CS2, i.e., cross-

section with and without fully enclosed noise barrier are shown in Fig. 10. Note that the amplitude in Fig. 10 represent the original experimental amplitude and all amplitude and wind speed are non-dimensionalized. It can be observed that regarding to the regular truss section without noise barrier CS2, there exists two VIV lock-in ranges, i.e., within reduced wind speed of 2.6–3.2 and 5.4–7.1 whereas the enclosed section with barrier CS1 has only one outstanding lock-in range of 4.4–7.2. Due to the degradation of the enclosed barrier on aerodynamic performance, the VIV amplitude of CS1 dramatically increases compared with CS2. If the entire span is designed with CS1, the large response, as shown in Fig. 10, will exceed the safety criteria of VIV

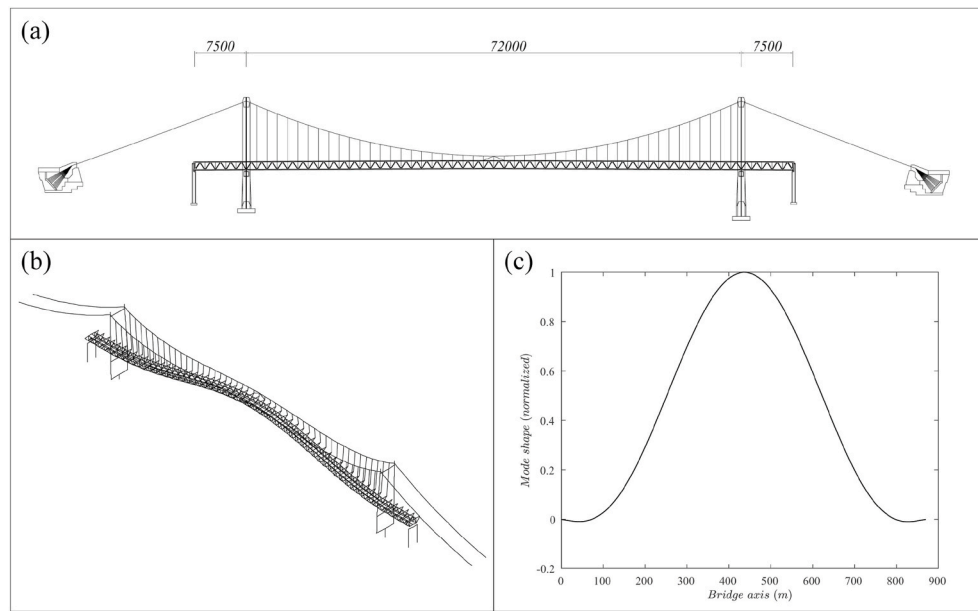


Fig. 6. (a) Layout of example bridge; (b) Sketch of 1st vertical mode shape of example bridge; (c) Normalized 1st vertical mode shape along bridge axis.

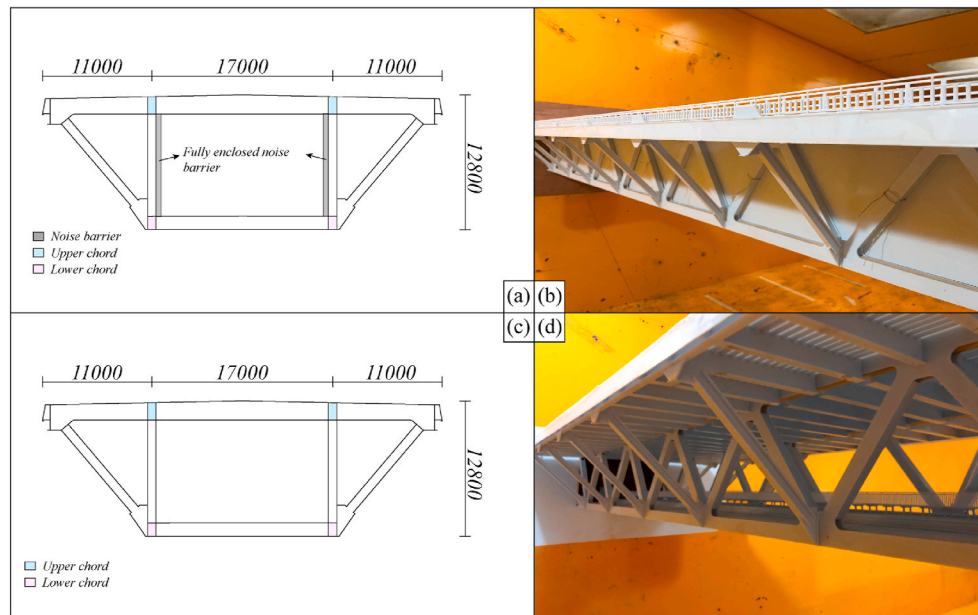


Fig. 7. Layout of cross-section and sketch of testing model (unit: mm): (a)–(b) cross-section with fully enclosed barrier (CS1); (c)–(d) cross-section without barrier (CS2).

amplitude defined by maximum acceleration of 1 m/s^2 according to Wind-resistant Design Specification for Highway Bridges of China (MOT, 2018) and significantly threat the operational safety of the bridge and vehicle and therefore calls for an inhomogeneous cross-section design, which enhances the necessity of this study. The aerodynamic parameters Y_1 , ϵ and H_1 identified using decay-to-resonance method and free decay vibration tests are shown in Fig. 11. As Y_1 and ϵ are associated with vortex-induced motion exclusively when the model is at lock-in, whereas H_1 is associated with aerodynamic damping exclusively when VIV is absent and therefore, the experimental Y_1 , ϵ and H_1 are not continuous against wind speeds in Fig. 11. In addition, when the reduced wind speed exceeds 7.5, both cross-sections show negligible VIV and thus no tests are performed for parameter identification beyond 7.5. It can be observed that out of lock-in, H_1 remains negative, which reveals

that the air-induced damping is hindering the motion of the model. The magnitude of H_1 on CS2 is apparently greater than CS1 which indicates that the enclosed barrier on CS1 yields lower air-induced damping and thus make CS1 more unfavorable with respect to VIV-resistance behavior. At lock-in, Y_1 remains positive, which suggests that the vortex-induced force is promoting the motion of the model. The magnitude of Y_1 on CS1 is generally greater than CS2, indicating that CS1 yields higher vortex-induced force and is aligned with the greater amplitude observed in Fig. 10. ϵ is a parameter coupled with Y_1 which mathematically represents the limited-amplitude characteristic of VIV and a higher ϵ usually yields a lower amplitude. Fig. 11(c) shows that the ϵ of CS2 is globally higher than CS1 which is aligned with the lower amplitude of CS2 observed in Fig. 10.

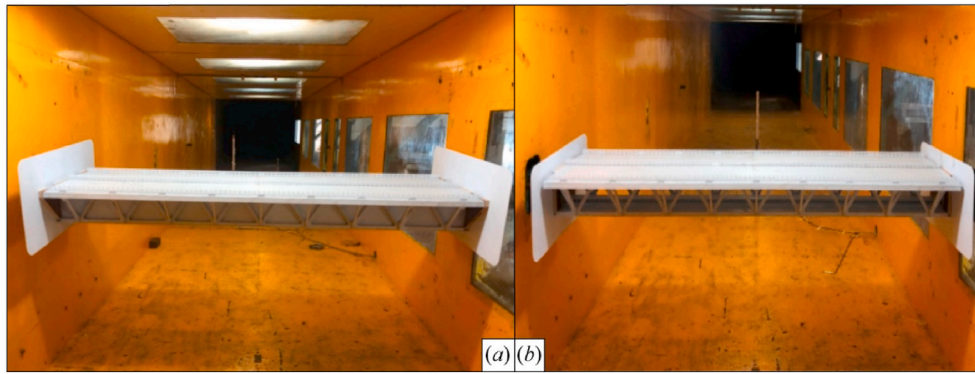


Fig. 8. Section model for wind tunnel test: (a) cross-section with enclosed barrier (CS1); (b) cross-section without barrier (CS2).

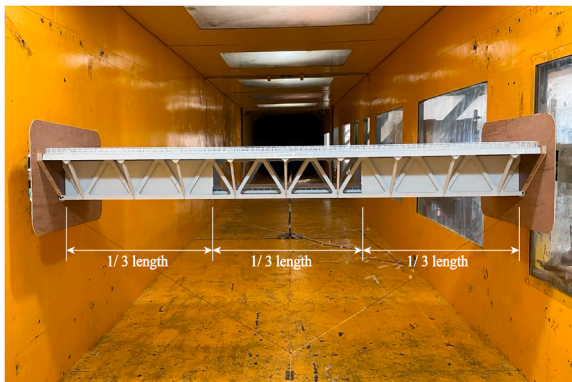


Fig. 9. Validation model with inhomogeneous cross-sections for wind tunnel test.

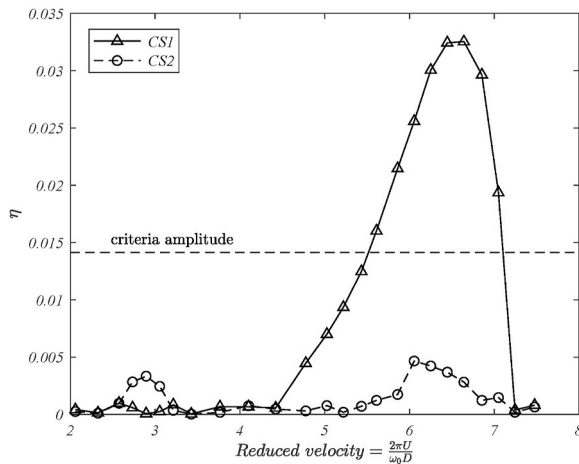


Fig. 10. Experimental VIV amplitude of sectional models against reduced wind speed.

4.2. Validation results

VIV amplitude of the validation model with inhomogeneous cross-sections is obtained via wind tunnel test, and given the aerodynamic parameters of CS1 and CS2 identified in Section 4.1, the proposed model described by Eq. (14) is readily to predict the VIV of the validation model. The comparison of experimental and predicted results is illustrated in Fig. 12 and the results of CS1 and CS2 are also displayed for discussion. Experimental results show that compared with CS1, both the

width of lock-in zone and the peak amplitude in the inhomogeneous results are reduced to a certain extent. The starting velocity of lock-in is reasonably delayed from 4.4 to 4.8 due to the additional air-induced damping caused by CS2 in the inhomogeneous model within this velocity range. Compared with the results of CS2, the first lock-in zone within reduced velocity 2.6–3.2 is absent and peak amplitude over the entire wind speed range significantly increases in the inhomogeneous results. It reveals that the VIV performance of inhomogeneous model is, as anticipated, situated between the results of CS1 and CS2. The predicted results using proposed model show reasonable agreement with the experimental data, which indicates that the proposed model can effectively reflect the essential aerodynamic mechanics of VIV for inhomogeneous cross-sections. For discussion purpose, the testing velocity is divided into three zones according to if the cross-section is at lock-in or not, i.e., zone I, II and III. In velocity zone I, only CS2 is at a minor lock-in and CS1 shows no significant VIV. In zone II, only CS1 is at lock-in. In zone III, both CS1 and CS2 are at lock-in. It can be observed that the proposed model performs well in zone I that it successfully predicts the vanishing of the first minor lock-in zone (2.6–3.2), which was observed on CS2. This phenomenon is probably because the additional air-induced damping brought by CS1 in the inhomogeneous model hinders the model's motion and breaks the critical condition for the development of VIV within 2.6–3.2. In zone II, the proposed model can reasonably predict the trend of amplitudes but somehow overestimate the starting velocity of lock-in. In zone III, the predicted results show favorable agreement with the experimental ones while still slightly underestimates the peak amplitude. The proposed model can also give satisfactory prediction of ending velocity of lock-in. One reasonable explanation for these deviations above is that the proposed model does not consider the effect of three-dimensional flow at the transition point between CS1 and CS2. Local flows at the transition point could be complex and inevitably change the aerodynamics of nearby segments, which could impose inaccuracy in the predicted results. On the other hand, the proposed model ignores the correlation of vortex-induced force in span-wise, i.e., all forces are considered as synchronous. In fact, the vortex-induced force can be out of phase if the span is long enough (Sun et al., 2019; Willden and Graham, 2001; Zhou et al., 2017) and thus might cause certain deviations in the modeled results. Another reason may cause errors in the proposed model is the insufficient data near the critical velocity of lock-in. Refining the resolution of identified parameters near the critical velocity could potentially improve the prediction, particularly near the starting velocity of lock-in. Nevertheless, the proposed approach could sensibly predict the essential VIV of the validation model with inhomogeneous design and can be considered as valid for further application on the prototype bridge.

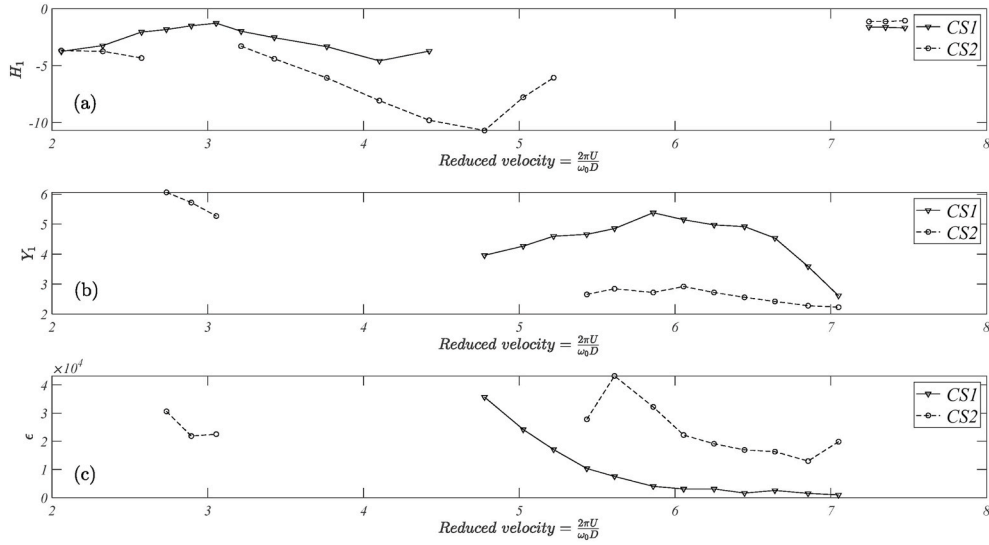


Fig. 11. Identified aerodynamic parameters H_1 (a), Y_1 (b) and ε (c) of two cross-sections via wind tunnel test.

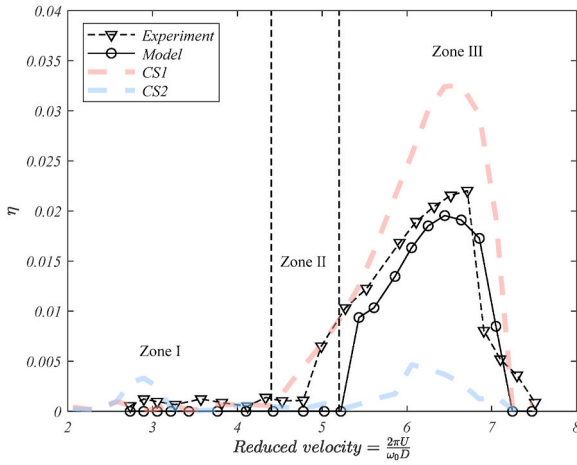


Fig. 12. Comparison of experimental and predicted results of inhomogeneous model.

4.3. Vortex-induced vibration of prototype bridge

Given the identified parameters of CS1 and CS2 and the well validated results in Section 4.2, the proposed integrated model is readily to deploy on the prototype bridge with inhomogeneous section design (33% CS1 + 34% CS2 + 33% CS1) and evaluate its global VIV performance. The VIV amplitude of the prototype bridge against wind velocity is demonstrated in Fig. 13. The VIV amplitude of the prototype bridge with only CS1 (full span with barrier) and only CS2 (full span without barrier) are also shown for comparison. It can be seen that the global VIV amplitude of inhomogeneous design is apparently increased by up to 140% compared with the result of only CS2 but still significantly smaller than the results of only CS1. Though the deployment length of CS1 (66% of the span) is greater than that of CS2 (34% of the span), the global amplitude of the inhomogeneous design is still closer to the results of only CS2. This phenomenon is due to the impact of mode shape in which the peak displacement occurs near the mid-span, as displayed in Fig. 6(c) and therefore gives the cross-section near the mid-span a magnification effect on global VIV amplitude. As a result, the global VIV amplitude is dominated by the cross-section near the mid-span even if its deployment length is not the most prominent. It is interesting to notice that the first minor lock-in of only CS2 within reduced velocity of 2.6–3.2 disappears

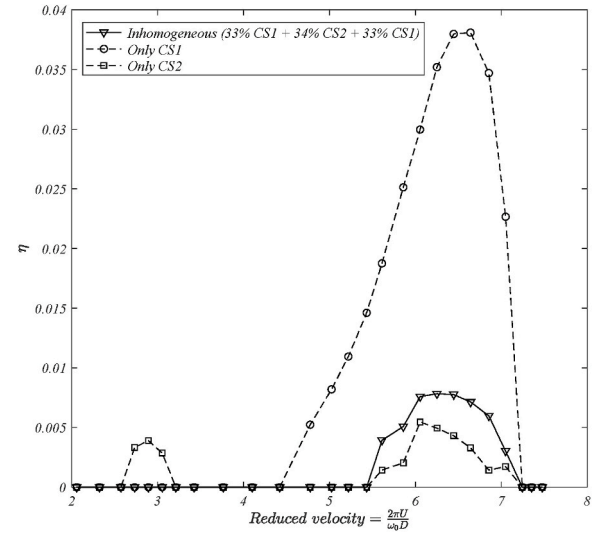


Fig. 13. VIV of prototype bridge with inhomogeneous cross-section design.

in the inhomogeneous results due to the additional air-induced damping caused by CS1 in the inhomogeneous design which increases the global damping and break the critical condition of VIV development. In general, the presence of enclosed noise barrier in the inhomogeneous design degrades the VIV performance of the bridge by increasing the peak amplitude up to 140%, but unexpectedly ameliorates the VIV behavior in low wind speeds by eliminating a minor lock-in region.

4.4. Effect of cross-section deployment

One premise task in engineering practice regarding to inhomogeneous cross-section design is to acquire the effect of different cross-section deployments on global VIV performance and possibly optimize the deployment to reach a balance between VIV and conceptual requirement. Regarding to the specific bridge example in this study, a long deployment of enclosed noise barrier may enable the bridge a better noise-cancelling performance, but it will dramatically increase the global VIV amplitudes. Therefore, it is necessary to investigate various lengths of barrier deployment and potentially choose a balanced plan. Two typical scenarios of barrier deployment will be discussed respectively, i.e., symmetrical deployment and asymmetrical

deployment.

4.4.1. Symmetrical deployment

Two cross-sections with enclosed noise barrier (CS1) and without enclosed noise barrier (CS2) are deployed symmetrically along the bridge. The one-sided deployment length of CS1 is increased from 0% to 50% and the rest of the span is deployed with CS2. Note that 50% indicates the entire span is deployed with CS1 and 0% indicates the entire span is deployed with CS2. Various deployments are investigated using the proposed approach and the results are shown in Fig. 14. It can be observed that when the one-sided deployment length of CS1 is below 20%, the VIV amplitudes are almost identical to those of 0%, which indicates that effect of inhomogeneous design on VIV is negligible when CS1 is only deployed near two bridge ends because the vertical displacement of mode shape near two bridge ends are quite limited. When the one-sided deployment length of CS1 increases from 30% to 50%, the minor lock-in region disappears whereas the non-dimensional VIV amplitude in the outstanding lock-in region rapidly grows from 0.005 to 0.038 due to the magnification effect of mode shape. It reveals that it becomes more sensitive to deploy CS1 near the mid-span of the bridge where the vertical displacement of mode shape is relatively large. It can be also noticed that the starting velocity of the outstanding lock-in is reduced from 5.4 to 4.4 with the increasing of deployment length. A greater deployment length of CS1 will not only enlarge the amplitude but also makes the lock-in velocity range wider. However, it shows that the predicted starting velocity of lock-in is not continuous which is probably due to insufficient resolution of sectional aerodynamic parameters against wind speeds. In the future, if one aims to acquire a better and smoother variation of lock-in velocity, the resolution of aerodynamic parameters should be carefully refined near the critical velocity of lock-in. Fig. 15(a) illustrates peak amplitude η_{peak} of VIV against the one-sided deployment length. It shows that the peak amplitude first grows slowly when the one-sided deployment length is under 25% then increases rapidly when the deployment length is over 25%, which is due to the magnification effect of mode shape.

It is interesting to discuss the potentially most suitable deployment plan of CS1 and CS2. An inherent conflict lies between the deployment length of enclosed noise barrier, i.e., CS1 and VIV amplitude. On one hand, noise cancelling calls for the increasing of the deployment length of noise barrier while on the other hand, VIV amplitude requires to shorten the noise barrier length. To evaluate the potentially most suitable deployment length regarding to the balance of noise cancelling and VIV, a target function represents the ratio of peak amplitude and noise cancelling performance is defined as:

$$V = \eta_{peak} / \text{noise cancelling performance} \quad (29)$$

The most suitable deployment length should reduce the peak VIV amplitude as much as possible while simultaneously maximize the noise cancelling performance, which is equivalent to minimize the target

function V . If we roughly assume that the noise cancelling performance is linearly increased with the deployment length of the barrier, which is reasonable to a certain extent because a longer barrier often yields better noise cancelling performance, V could be further reduced as:

$$V = \eta_{peak} / \text{deployment length} \quad (30)$$

The deployment length yielding the minimal V could be regarded as a suitable barrier deployment length which reaches a favorable balance between noise cancelling and VIV amplitude. The variation of V against the one-sided deployment length is illustrated in Fig. 15(b). With the increasing of deployment length of barrier (CS1), the curve of V first drops quickly and reaches a minimal value, then rises rapidly. A short deployment length of barrier ($\leq 10\%$) yields a high V due to the poor noise cancelling performance even though the peak amplitude of VIV is relatively low. When the deployment length is 28%, V meets the minimal point which represents a potentially most suitable deployment length of CS1. When the deployment length is over 28%, V rises rapidly because the peak amplitude of VIV η_{peak} grows fast due to the magnification effect of mode shape. To sum up, the one-sided deployment length of 28% seems to yield the most favorable balance between noise cancelling and VIV amplitude. Note that the result above is based on the linear relationship between noise cancelling performance and deployment length of noise barrier due to insufficient studies on this specific issue. In the future, if the explicit relationship is given, the result could be refined to provide a better suggestion on the most suitable deployment length.

4.4.2. Asymmetrical deployment

In actual engineering practice, inhomogeneous design could be asymmetrical, and it is interesting to investigate the effect of asymmetrical cross-section deployment on the global VIV. For the specific bridge case in this study, the enclosed noise barrier (CS1) is deployed from one bridge end and stretches to the other to investigate the VIV behavior under various lengths of asymmetrical barrier deployment. Global VIV with deployment length from 0 to 100% are evaluated and the results are shown in Fig. 16. It shows that when deployment length is within 0–30%, the VIV amplitude barely changes whereas when deployment length reaches 40%–70%, the VIV amplitude rapidly increases. Moreover, the minor lock-in region within reduced velocity range 2.6–3.2 disappears when deployment length is beyond 30%. The width of the outstanding lock-in region also grows with the increasing of deployment length. Fig. 17(a) illustrates the variation of peak amplitude η_{peak} against deployment length. It indicates that η_{peak} first barely increases and then rapidly rises and stops growing at non-dimensional amplitude of 0.038 when the deployment length is over 75%. Similar with the previous section, to investigate the most suitable asymmetrical deployment length of barrier, the target function of V is evaluated. The variation of V against deployment length is illustrated in Fig. 17(b). It reveals that the minimum V occurs when deployment length is about

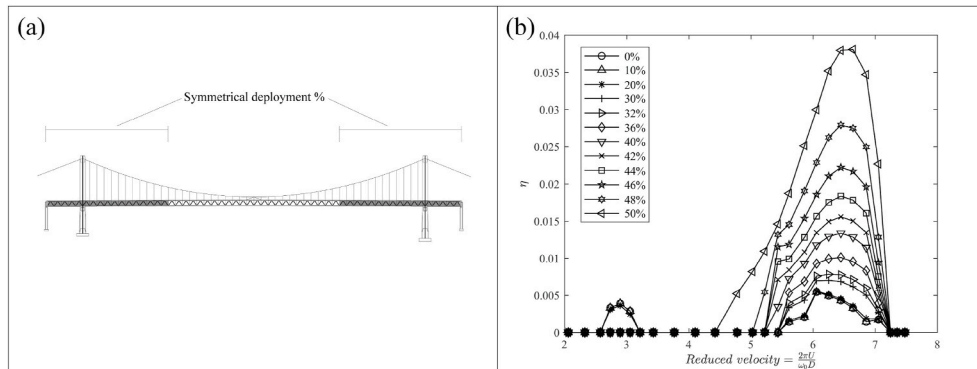


Fig. 14. VIV of various symmetrical deployments: (a) sketch of symmetrical deployment; (b) global VIV amplitude against reduced velocity.

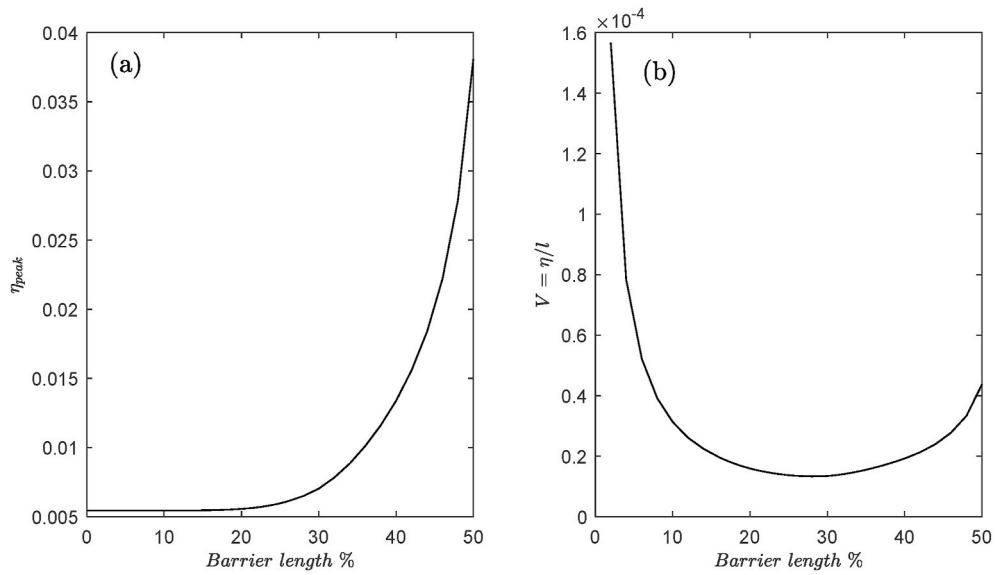


Fig. 15. (a) Peak amplitude of VIV η_{peak} against the one-sided deployment length of barrier; (b) target function V against the one-sided deployment length of barrier.

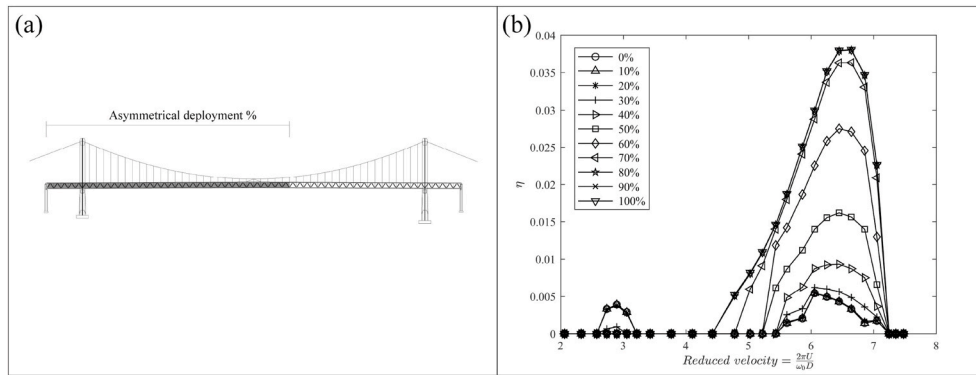


Fig. 16. VIV of various asymmetrical deployments: (a) sketch of asymmetrical deployment; (b) global VIV amplitude against reduced velocity.

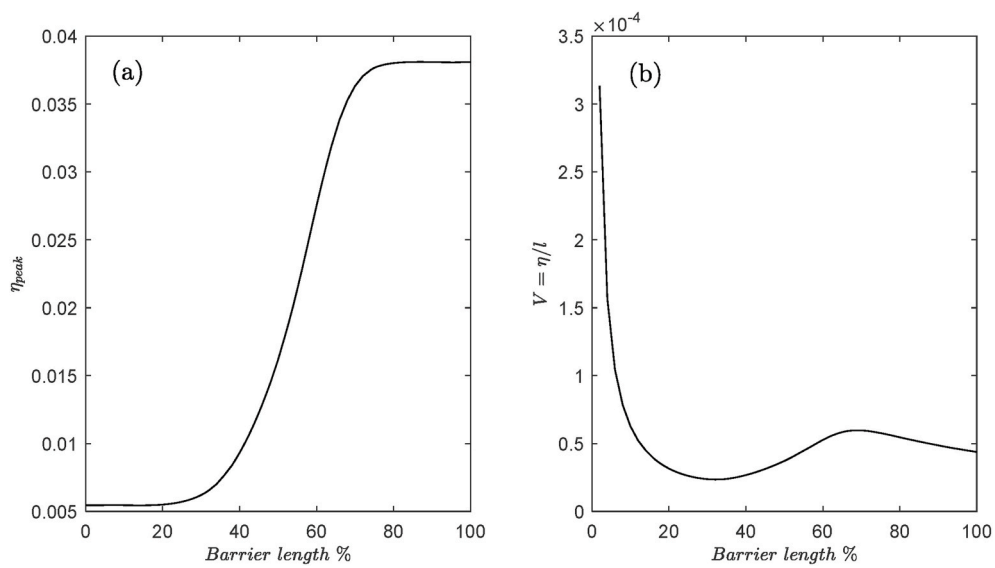


Fig. 17. (a) Peak amplitude at mid-span against deployment length; (b) V against deployment length.

32%, which reaches the most favorable balance between noise cancelling and VIV amplitude, whereas the maximum V occurs when deployment length is 70%, which yields a poor balance of barrier deployment and should be potentially avoided.

5. Conclusion

In this study, an integrated approach which combines Scanlan's semi-empirical nonlinear vortex-induced force model with a linear air-induced damping model is presented as an alternative to investigate the vortex-induced vibration of long-span bridge with inhomogeneous cross-sections. An urban long-span bridge partially enhanced with enclosed noise barrier to reduce noise pollution for nearby inhabitants near bridge ends is regarded as an example. The aerodynamic parameters of each cross-section within and out of lock-in region are identified via sectional model wind tunnel tests using decay-to-resonance or free decay vibration method and implemented in the proposed approach. A rigid segment with inhomogeneous cross-sections is designed as a validation model to verify the feasibility of the proposed approach. The experimental and predicted results are carefully compared. The effect of inhomogeneous design on global VIV amplitude is investigated and various cross-section deployments are investigated and discussed.

The proposed approach can reflect the essential aerodynamic mechanics of bridge with inhomogeneous cross-sections and gives reasonable predictions compared with experimental results whereas still overestimates the starting velocity of lock-in and slightly underestimates the peak amplitude. Such deviations might be caused by the ignorance of three-dimensional effect of local flow at the transition point and spanwise correlation of vortex-induced force. Span-wise inhomogeneous design may have all-around impacts on VIV performance compared with conventional homogeneous design including alteration in lock-in speed, lock-in regions, and peak amplitude. For the specific case in this study, the inhomogeneous design with the cross-section of fully enclosed noise barrier generally degrades the global VIV performance of the bridge by increasing the peak amplitude up to 140%, but unexpectedly ameliorates the VIV behavior in low wind speeds by eliminating a minor lock-in region. Mode shape has magnification effect on global VIV for inhomogeneous design. The inhomogeneity in cross-section shows negligible influence at locations with limited modal displacement whereas significantly affects the global VIV at locations with large modal displacement. Due to the magnification effect of mode shape, cross-section with worse VIV performance should be deployed away from locations with large modal displacement, e.g., mid-span for 1st vertical mode. For the specific case in this study, it is found that the deployment length of noise barrier being 28% and 32% for symmetrical and asymmetrical scenarios, respectively, yields the most suitable deployment plan, i.e., meeting the best balance between VIV amplitude and noise cancelling performance.

In the future, comprehensive factors such as span-wise correlation and local flow effect at the transition point should be reasonably taken into consideration. The proposed approach should also be validated against a full bridge test to examine the potential discrepancies which may be caused by interaction of multiple elastic modes. The inherent nature of amplitude dependency and nonlinearity in air damping is simplified herein to reduce the complexity of the problem. More research on the effect of nonlinear amplitude-dependency damping on inhomogeneous cross-section should be investigated to improve the present integrated model. It should be noted that the most suitable deployment length of noise barrier in this study is based on the linear relationship between noise cancelling performance and deployment length of noise barrier due to insufficient studies on this issue. If the explicit relationship is given, the result could be refined to provide a better suggestion on the most suitable deployment length in the future.

CRedit authorship contribution statement

Junzhi Pan: Writing – original draft, Software, Formal analysis,

Experiments, Validation. **Zilong Ti:** Conceptualization, Writing – review & editing, Supervision. **Yubing Song:** Experiments, Data curation. **Yongle Li:** Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work presented in this article.

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